

Design of 3D-Printed Soft Sensors for Wire Management and Customized Softness

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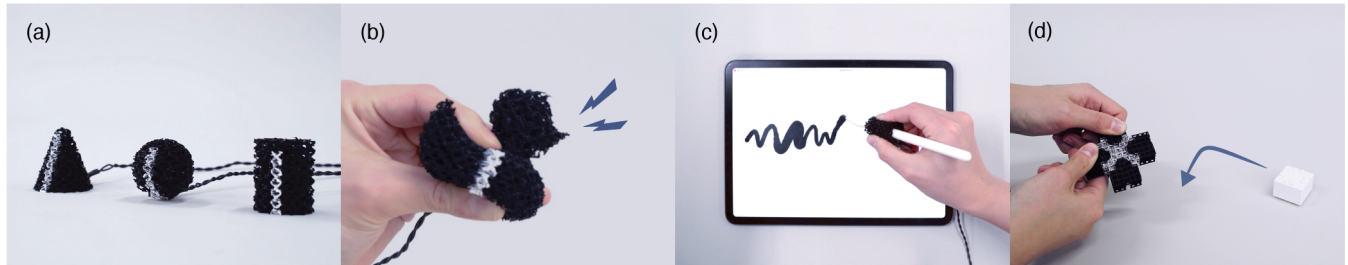


Figure 1: (a) Sensors with various three-dimensional shapes, (b) Responsive Soft Duck, (c) Interactive Pen Grip, (d) Soft Controller

ABSTRACT

Although the interaction capability of soft sensors has been explored in the field of HCI, they are found to have limitations in shape, wiring management, and softness due to the characteristics of the material. In this paper, we propose a design of 3D-printed resistive soft sensors that can detect deformation by creating internal lattice structures with flexible conductive materials. The magnitude of deformation can be detected by measuring the resistance between a pair of electrodes. By adjusting the parameters of the lattice structure, a soft sensor with high flexibility of shape design and adjustable local softness can be realized. In addition, by inserting non-conductive structures into the sensor structure, external wires can be consolidated in one place. We present a design and fabrication pipeline, and investigate the fundamental characteristics of several patterns of sensors. We also show several applications that demonstrate the feasibility of the proposed method.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

3D printing; soft sensor; fabrication; deformable user interfaces

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1 INTRODUCTION AND RELATED WORK

Soft interfaces, such as textiles [6], elastomer [8], and foams, are flexible and lightweight that can be easily integrated into daily life. They provide an environment to achieve more natural interactions than rigid interfaces. In line with this trend, a body of research has been focused on methods to realize more natural and organic interactions by incorporating soft objects into interfaces for many electronic devices composed of hard parts [1].

In designing such soft interfaces, deformation is often used as a direct input method to the computer. Many soft sensors are based on combining materials, such as wrapping a hard sensor with a soft material. On the other hand, soft sensors composed entirely of soft materials have been designed by devising material properties and structures. This type of soft sensor can detect complex deformations that are impossible with rigid sensors, thus detecting various gestures and realizing expressive interfaces. Conductive foam is particularly suitable for making three-dimensional sensors. *Foam-Sense* [4] and *Foamin* [7] leverage a soft sensor using conductive foam, which has the advantages of not losing the freedom of deformation and being easy to increase in size. However, it is difficult to fabricate the conductive foam by 3D printing. Thus, we have to use commercially available foam of uniform material, making it challenging to fabricate complex shapes and change the stiffness. In addition, the sensor will produce the same signal value regardless of the type of gesture such as pushing, twisting and shearing. The

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number of distinct gestures is limited because it is difficult for a single sensor to recognize gestures with the same signal value.

A soft sensor using multi-material 3D printing allows us to create complex shapes and freely design the internal structure. This method uses conductive and non-conductive filaments to design a sensor. *MetaSense* [3] is a capacitive sensor that uses a lattice-like structure for an object's internal structure. This soft sensor measures the capacitance change caused by the deformation of the unit cell. It can detect continuous deformation, but there is little information on how to incorporate it into a three-dimensional object. In addition, the wiring is complicated when there are many conductive cells inside.

As an approach using flexible conductive filaments, a soft gripper with a built-in sensor was also proposed [2]. The resistance change was measured by the change in resistance between the two electrodes due to deformation. However, the amount of deformation cannot be large because the sensor is laminated in a dense state. The sensor part is printed in a line; therefore it is not three-dimensional and can only detect deformation caused by bending. If a piezoresistive sensor is not used, the two electrodes must be placed face-to-face, so the position of the external wires cannot be flexibly located.

To address these issues, we propose a novel method to fabricate a soft sensor with a high degree of freedom in selecting the three-dimensional shape and wiring position by using a conductive flexible filament as the material using a 3D printer. In this method, the stiffness of the sensor can be changed even when using a conductive flexible filament by designing the internal structure, which is inspired by the research on changing the hardness by structure [5]. By using a lattice structure as the internal structure and changing the type and size of the lattice structure, we realized a three-dimensional soft sensor that can tune the softness. In addition, by using a non-conductive flexible filament as an insulator for the structure, we designed a conductive path that can be routed back and forth from one position to another. Thus, we can localize the wiring into one place to increase the ease of wire management. Furthermore, the soft sensor fabricated by this method can act as an output (single sensor) and can be easily incorporated into a three-dimensional object.

In this paper, we describe the system overview, design, and fabrication method of the proposed sensor and summarize the results of experiments to evaluate the performance of the sensor. Finally, an example implementation of an application using a sensor is presented.

The contributions of this research can be summarized as follows.

- 3D printing a lattice structure with conductive flexible filaments to create a soft sensor having high flexibility of shape design,
- Increasing the wiring flexibility by designing patterning of conductive and non-conductive materials,
- The softness of the soft sensor can be changed by changing the type of lattice structure, lattice size, and thickness of the pillars.
- Implemented an application that took advantage of the three-dimensional shape and wiring and demonstrated its applicability.

2 OUR APPROACH

2.1 Basic Design

A conductive flexible filament has a material property in which its resistance value changes when stress is applied. The basic principle of detecting deformation is to obtain the change in the resistance value due to the conductive part's deformation. In this sensor, a lattice structure was adopted as the internal structure to increase the amount of deformation caused by pushing, achieving softness, and a change in resistance sufficient to be utilized as a sensor.

The lattice structure can be modified in terms of deformation and softness by changing its unit lattice's size, type, and pillar thickness. Because the structure of the soft sensor as a whole can be changed depending on the part, it is possible to fabricate a soft sensor with different mechanical responses locally.

In order to acquire deformation by resistance measurement, two electrodes need to be attached in such a way that the deformed region of the conductive part is sandwiched from both sides. In this case, if the wires come in contact with the area, the softness is lost, and the interaction is affected (Fig. 2 (a)). Therefore, in this sensor, we adopted a structure that can change the position of the electrodes at two points by separating the conductive part by the non-conductive part, as shown in Fig. 2 (b). In the following, this non-conductive part will be referred to as the separator.

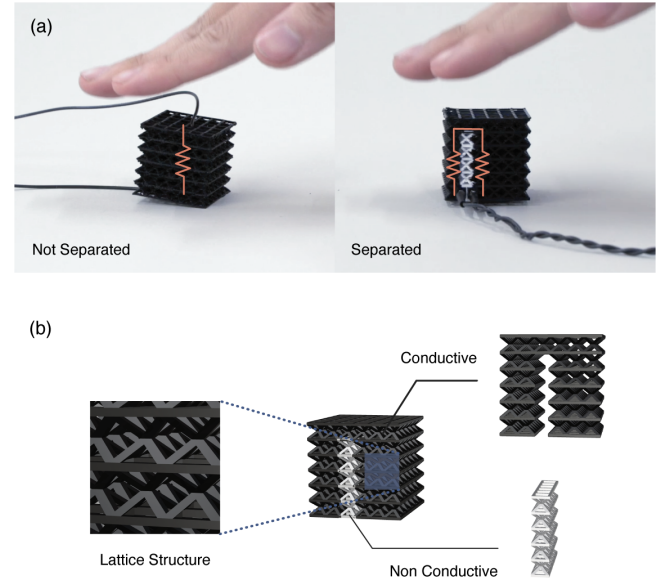


Figure 2: (a) Comparison of wiring with/without separator, (b) Configuration of the sensor

2.2 Fabrication

This sensor is fabricated using the following procedures: (1) 3D modeling, (2) 3D printing, and (3) thermal press fitting (Fig. 3).

3D modeling

Create the 3D model data of the sensor by following the steps below:

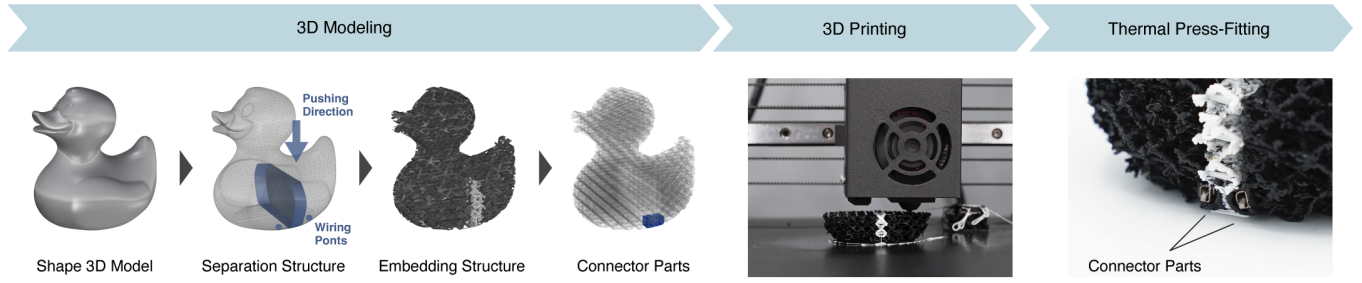


Figure 3: Design flow of the sensor

- (1) Prepare a 3D model to be the shape of the sensor.
- (2) Create a separation structure based on the pressing direction and wiring points.
- (3) The type and parameters of the lattice structure are determined and embedded into the model.
- (4) Attach the connector parts to the wiring points.

The separator should be placed parallel to the direction of the push-in and across the two wiring points. Further studies are to be carried out to understand the difference in the sensor function, depending on the angle and shape of the separator. The separator is the part that will be printed with a non-conductive filament, and the remaining part except for the separator is printed with the conductive filament.

The connector parts are designed as rectangles with hollowed out cylinders, and placed in contact with the conductive parts to be able to thermally press-fit the contact pins as shown in the right of Fig. 3. With the arrangement in the right of Fig. 3, the two wiring points can be brought closer together than the unit grid size, so the wire organization can be simplified.

3D printing

Slice the generated 3D model and print it out using a multimaterial 3D printer. Using conductive and non-conductive flexible filaments with equal Shore hardness makes it easier to make the entire softness of the soft sensor uniform, as the same internal structure will be equally soft when printed.

Thermal press-fitting.

Attach the contact pin to the connector part of the sensor after printing by heat-press fitting.

3 FABRICATED SENSORS AND THEIR CHARACTERISTICS

We printed a soft sensor with the shape and the structure shown in Fig. 4 (a) using the proposed method. The thickness of the pillars and frames was fixed at 0.8mm, which is twice the nozzle diameter, and three patterns of 30mm per side cubic sensors were printed by changing the type of lattice structure and the size of the unit lattice. The three patterns are body-centered cubic lattice without pillars (BCC-np) with unit lattice size of 5mm and 6mm, and body-centered cubic lattice without frames and pillars (BCC-nf) with unit lattice size of 5mm.

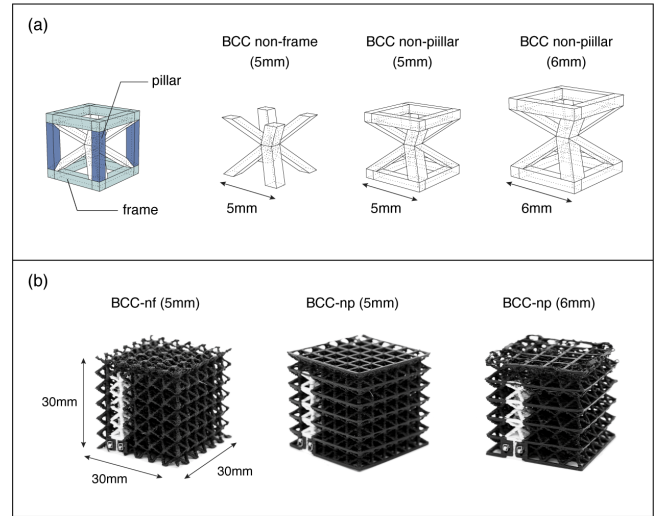


Figure 4: (a) Unit cell of lattice structure, (b) Three types of different printed soft sensors

NinjaTek's Eel filament was used as the conductive thermoplastic polyurethane (TPU) filament, and 3DXFLEX's Flexible TPU Filament was used as the non-conductive TPU filament. Both filaments have a Shore hardness of 90A.

We measured the compressive load and the electrical resistance of sensors by incrementally compressing the test cubes with 2.0mm. We waited for 30 seconds for each compression to stabilize the measurement value.

Fig. 5 (a) shows the relationship between the compression distance and (a) the compressive load of the fabricated sensors. The curve of the compressive load can be customized using different type of lattice structures: In the case of BCC-nf, it shows almost linear response, on the other hand, in the case of BCC-np, it indicates step-like response. Also, the steepness of the curve seems to be adjusted by the lattice size.

Fig. 5 (b) shows the characteristics of the electrical resistance of the sensors. They are all in the appropriate ranges (several kΩs) for the microcontroller to detect the deformation.

Even with the same model size, the range of resistance can be customized by changing the pattern of the lattice.

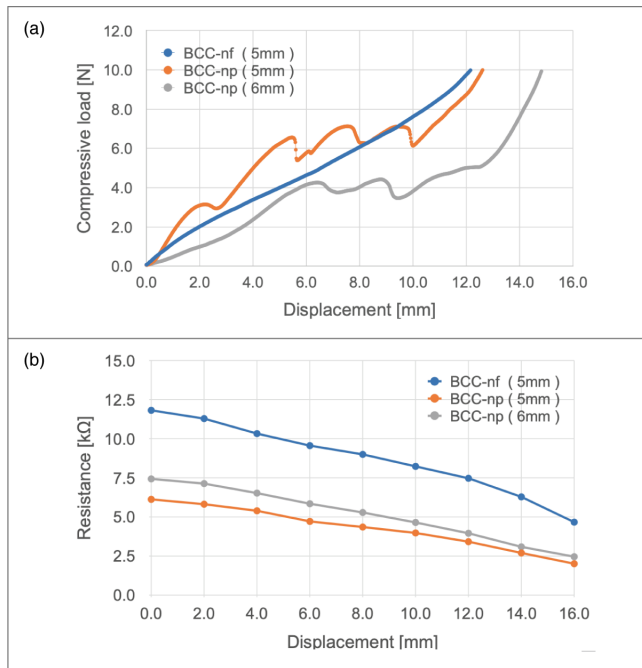


Figure 5: Relationship between the compression distance and (a) the compressive load, (b) the electrical resistance of the printed sensors.

4 APPLICATION EXAMPLES

4.1 Responsive Soft Duck

We fabricated a soft toy with the shape of a duck shown in Fig. 1 (b). When the body part is pushed in, it detects the deformation and makes a sound in response. The pitch of the duck's quack changes depending on the degree of push-in.

4.2 Interactive Pen Grip

The shape and softness of the pen grip can be customized to fit the shape of the pen and the user's hand grip strength. We implemented a soft pen grip to detect hand grip strength shown in Fig. 1 (c). When writing on the screen, the thickness of the line can be changed intuitively depending on the amount of push-in as shown in Fig. 6 (c).

4.3 Soft Controller

We created a soft controller that can detect push-in in four directions by placing four sensors independently. Taking advantage of the freedom of electrode placement, the wiring of the four sensors is grouped so that it can be taken out from the center of the backside, as shown in Fig. 6 (c). Here, we used it as a controller for toio, as shown in Fig. 1 (d). It is implemented so that it moves in each direction depending on the part pushed in, and the speed changes depending on the amount of pressing.

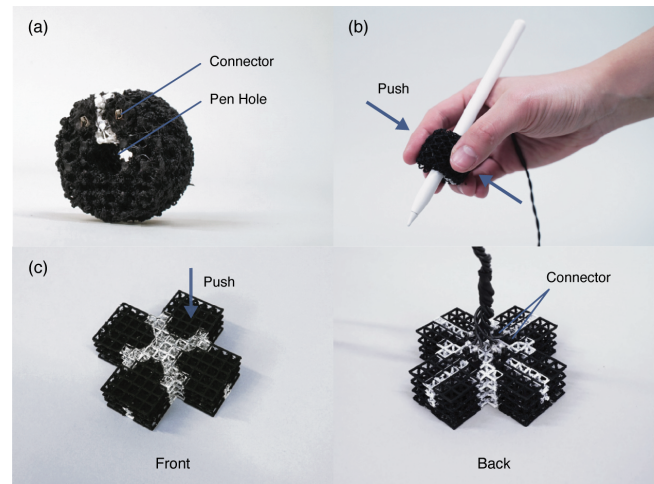


Figure 6: (a) Details of Pen Grip, (b) Push-in action with Pen Grip, (c) Details of Soft Controller

5 CONCLUSION

We proposed a new soft sensor formed into any desired shape using a 3D printer. We used a conductive TPU filament as the conductive material and placed a non-conductive TPU filament as a separator to increase the flexibility of the wiring position. The softness of the sensor can be changed locally by changing the internal structure. In the future, we plan to improve the design of the separator as described in the Basic Design and clarify the relationship between types of unit cell and the sensor feedback. We are also considering to explore structures that can detect deformations other than compression and develop tools to simplify the sensor design.

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